MIL-D-60027A (AR)
19 May 1978
SUPERSEDING
MIL-D-60027 (MU)
15 April 1964

MILITARY SPECIFICATION

DETONATOR, ELECTRIC, M69

PARTS FOR

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts for one type of initiating device designated as Detonator, Electric, M69.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIL-A-48076 - Ammunition, Standard Quality Assurance Provisions

STANDARD

MILITARY

MIL-STD-105D - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-1169 - Packaging, Packing and Marking, Inert Ammunition Components

DRAWINGS

8857199 - Detonator, Electric, M69

FSC: 1390

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 Parts.-The parts shall comply with all requirements specified on Drawing (Dwg. 8857199) and with all requirements specified in applicable specifications.

3.2 Parts.-The parts shall comply with all requirements specified on Drawing (Dwg.) 8857199, and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts shall be fabricated and finished in a thorough workmanlike manner. The parts shall be free of burrs, chips, cracks, sharp edges, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

3.4 First Article Inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article sample by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are part of the detailed specification.

4.2 Classification of Inspections.-The following types of inspections shall be conducted on this item:

a. First article inspection of detonator parts
b. Production Quality Conformance inspection of detonator parts

4.3 First Article Inspection.-These inspections are to be conducted to assure the quality of the detonator parts prior to the start of regular production.

4.3.1 Submission.-The manufacturer of the detonator parts, prior to initiation of regular production, shall ship to a Government approved facility designated by the Contracting Officer for evaluation, an initial production sample consisting of twenty-five (25) sets of parts for evaluation in accordance with the provisions of 4.3.2.

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4.3.3 Rejection.-(See MIL-A-48078 (PA)).

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with lot formation provisions of MIL-A-48078.

4.4.2 Examination.-One hundred percent examination shall be performed for all critical defects and major defects when so specified. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein using sampling plans of Standard MIL-STD-105 Level II. All non-conforming material shall be rejected (See MIL-A-48078).

NOTE: Critical defect reporting.-A report of all critical defects found, number of parts or assemblies inspected for critical defects, and corrective action taken, if any, shall be compiled monthly for the duration of the contract and copies forwarded to the Contracting Officer and Commanding General, US Army Armament Research and Development Command, ATTN: DRDAR-LCR-E and DRDAR-QA-A-F, Dover, NJ 07801.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60027A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Detonator, Electric, M69, Parts for				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cup Dwg 8857193 Examination of defects	25	(a)	3.2/3.3	4.4.2.1
	Pin, After Coating Dwg 8857194 Examination of defects	25	(a)	3.2/3.3	4.4.2.2
	Plug Dwg 8857195 Examination of defects	25	(a)	3.2/3.3	4.4.2.3
	Plug Assembly Dwg 8857196 Examination of defects	25	(a)	3.2/3.3	4.4.2.4

NOTES: (a) to be performed at contractor's plant.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60027A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Cup				8857193
					NEXT HIGHER ASSEMBLY 8857199
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Cup cracked or split		100%	3.2	Visual
Major					
101.	Thickness thru bottom		0.40%	3.2	Gage
102.	Outside diameter minimum (Min)		0.40%	3.2	Gage
103.	Inside diameter		0.40%	3.2	Gage
104.	Radius, min		0.40%	3.2	Gage
Minor					
201.	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60027A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Pin, after coating				8857194
					NEXT HIGHER ASSEMBLY
					8857196
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Thickness of coating		0.40%	3.2	Gage
102.	Diameter of pin at required dimension (after removal of coating) (Note 3 of dwg. 8857194)		0.40%	3.2	Gage
Minor					
201.	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60027A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Plug				8857195
					NEXT HIGHER ASSEMBLY
					8857196
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Total length		0.40%	3.2	Gage
102.	Large outside diameter, min.		0.40%	3.2	Gage
103.	Small outside diameter, min.		0.40%	3.2	Gage
104.	Small diameter of pin hole, max.		0.40%	3.2	Gage
105.	Large diameter of pin hole, max.		0.40%	3.2	Gage
106.	Radius outside diameter at large end		0.40%	3.2	Gage
107.	Finish improper		0.40%	3.2	Visual
Minor					
201.	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60027A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Plug Assembly	1 OF 1	8857196
			NEXT HIGHER ASSEMBLY 8857197

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Resistance between pin and plug (Note 2 of dwg)		100%	3.2	Test 4.5.1
102.	Push test of pin		0.40%	3.2	Test 4.5.2
103.	Length pin protrudes		0.40%	3.2	Gage
104.	Pin not flush with end of plug		0.40%	3.2	Gage
105.	Separation of bridging point		0.40%	3.2	Gage
Minor					
201.	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

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4.4.3 Testing

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed therein, is described in the paragraph Inspection Method Column in the tables starting in paragraph 4.4.2.1. The contractor shall submit for approval inspecting equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test Methods and Procedures

4.5.1 Resistance between Pin and Plug.-The plug assembly shall be connected to the applicable test fixture specified in 4.4.4 and the voltage shall be applied for the time specified on the applicable drawing. Any assembly which fails to comply shall be classed defective.

4.5.2 Push test of Pin at 70 degrees Fahrenheit.-The plug assembly shall be assembled to the applicable test fixture specified in 4.4.4 and the load specified on the applicable drawing applied to the pin. Any assembly which shows evidence of movement between the pin and the plug shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking.-Packing and marking shall be in accordance with Specification MIL-STD-1169.

6. NOTES

6.1 Intended Use.-The detonator covered by this specification is intended for use in Fuze, Point Initiating Base Detonating, M509A1E1, and Fuze PI, BD, M530A1.

6.2 Ordering data.-(See 6.1 of MIL-A-48078) (PA).

6.3 Submission of Design Approval.-See 6.2 of MIL-A-48078. Submit equipment designs as required to Commander, ARRADCOM Hqts, Dover, NJ 07801, ATTN: DRDAR-QAA-F.

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Preparing Activity:
Army-AR

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-D-60027A Detonator, Electric, M69 Parts For

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